BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. In urea cycle, arginine splits into urea and ornithine by an enzyme:

(A) Arginase
(B) Dehydrogenase
(C) Ornithase
(D) Citrulase

Q2. Which vertebra in reptiles is modified for the rotational movement:

(A) Axis
(B) Atlas
(C) Sacral
(D) Thoracic

Q3. At the place of attachment of leaf with the shoot, a swollen part is called:

(A) Pit
(B) Cortex
(C) Pulvinus
(D) Cortex

Q4. Resting membrane potential of a neuron is:

(A) -50 mV
(B) -60 mV
(C) -70 mV
(D) -80 mV

Q5. Abscisic acid can be sprayed on tree crops to regulate:

(A) Leaf drop
(B) Shoot drop
(C) Pollen tube
(D) Fruit drop

Q6. Vehicles for transport of male gamete in land plant is :

(A) Pollen tube
(B) Pollen grain
(C) Vacuole
(D) Anther

Q7. Cell wall becomes thicker and pitted during cell:

(A) Maturation
(B) Elongation
(C) Differentiation
(D) Division

Q8. Copying of mRNA from DNA is called:

(A) Transduction
(B) Transdation
(C) Transformation
(D) Transcription

Q9. DNA polymerase III:

(A) Recognizes primer
(B) Recognizes primer
(C) Initiates DNA replication
(D) Unwinds DNA helix

Q10. Down syndrome is

(A) Trisomy 19
(B) Trisomy 18
(C) Trisomy 23
(D) Trisomy 21

Q11. Bombay phenotype is an example of:

(A) Pleiotropy
(B) Epistasis
(C) Probability
(D) Dominance

Q12. Primer for PCR contains about

(A) 05-07 bases
(B) 25-30 bases
(C) 10-20 bases
(D) 30-40 bases

Q13. One common type of vector is:

(A) Plasmid
(B) Lysosome
(C) Mitochondria
(D) Chromosome

Q14. The ultimate source of all changes is :

(A) Migration
(B) Mutation
(C) Genetic drift
(D) Selection

Q15. Overgrazing may lead to:

(A) Tundra
(B) Grassland
(C) Desert
(D) Taiga

Q16. Scum in eutrophication is formed by:

(A) Algae
(B) Virus
(C) Bacteria
(D) Fungi

Q17. Which of these diseases is caused due to nutritional deficiency:

(A) Diphtheria
(B) Osteoarthritis
(C) Arteriosclerosis
(D) Scurvy

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What is counter current multiplier mechanism?

(ii) Define excretophores Give their functions

(iii) Give the role of pyrogens.

(iv) What is negative geotropism? Give at least one example

(v) Write the composition of procuticle.

(vi) Give the structure of sarcoplasmic reticulum.

(vii) What is ovoviviparity? Give an example.

(viii) Draw and label the diagram of C.S. of seminiferous tubule

(ix) What is difference between climate and weather?

(x) What is grassland ecosystem? Give at least one example.

(xi) Define soil.

(xii) Draw the flow chart showing the formation of ASH and CO2 from dead plants.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Define nerve impulse.

(ii) Define nociceptors.

(iii) What do you know about cretinism?

(iv) How can you protect the baby against Rh incompatibility?

(v) What is MODY?

(vi) Write down two practical uses of DNA finger printing.

(vii) In birds, the female is heterogametic. How?

(viii) Discuss any two benefits of transgenic bacteria to promote health in plants.

(ix) Define and give examples of ex-vivo and in-vivo gene therapy.

(x) What are biogeochemical cycles?

(xi) Discuss role of decomposers in ecosystem.

(xii) Define food chain. Write an example.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) How light plays important role in plant growth?

(ii) How light plays important role in plant growth?

(iii) In what way mutation causes sickle cell disease?

(iv) Why replication always take place in 5' -> 3' direction?

(v) What do you know about Okazaki fragments?

(vi) Compare mitosis with meiosis.

(vii) Write symptoms of Down's syndrome.

(viii) Differentiate between homologous and analogous organs.

(ix) What are vestigial organs? Give one example.

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [4] What is Renal failure? Describe its cure.

(b) [4] What is cancer? Give its causes and effects

Q6.

(a) [4] Define joints. How are they classified? Explain.

(b) [4] Define succession. Explain xerosere in detail

Q7.

(a) [4] What is active membrane potential? Explain its major causing factors. Also draw a graph that shows changes in membrane potential from resting to active membrane potential.

(b) [4] Define Hardy-Weinberg theorem. Also explain the Hardy-Weinberg equations for calculating the frequencies of alleles and genotypes in populations at equilibrium.

Q8.

(a) [4] Explain the birth of twins in human beings:

(b) [4] Describe the assortment of alleles of two contrasting pairs of traits when followed in the same cross by giving one example.

Q9.

(a) [4] What is growth? Discuss different conditions for growth.

(b) [4] Define gene therapy, explain in which diseases ex-vivo gene therapy are needed.